Function Notation: When $f(x)$ Is Not Multiplication Answer Key

Algebra 1 — Functions

Quick Answers

1. 7	3. 20	5. 40	7. 0, 4
2. 23	4. 11	6. 4	8. —

Curriculum

Level: Algebra 1 — Functions
HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got 21: read f(3) as f times 3 and multiplied the whole rule by 3*
2. *If they got 115: read g(5) as g times 5 and multiplied the whole rule by 5*
3. *If they got 37: moved the outside factor 2 inside the parentheses and evaluated f(6) instead of doubling f(3)*
4. *If they got 33: distributed a factor of 3 over the rule, treating f(3) as 3 times f*
5. *If they got 28: split the input as if f distributed over addition, computing f(2) + f(3)*

1. $f(x) = x + 4$; find $f(3)$.

Solution: The 3 inside the parentheses is an input, so it replaces x everywhere in the rule. Substitution step: $f(3) = (3) + 4$. Arithmetic: $3 + 4 = 7$. $\boxed{7}$

Watch for: an answer of 21 means the student read $f(3)$ as $f \cdot 3$ and multiplied the whole rule by 3, giving $3(x + 4)$ at $x = 3$. Notice that this reading never uses the 3 as an input at all.

2. $g(x) = x^2 - 2$; find $g(5)$.

Solution: Replace x with 5: $g(5) = (5)^2 - 2$. Square first, then subtract: $25 - 2 = 23$. $\boxed{23}$

Watch for: an answer of 115 is $5(x^2 - 2)$ evaluated at $x = 5$ — the multiplication misreading again. The size of the gap is the tell: the correct output is smaller than the input squared, while the misread value is five times too big.

3. $f(x) = x^2 + 1$**; compute** $2f(3)$**.**
Solution: The 2 sits *outside* the function name, so it scales the output, and the only thing inside the parentheses is the input 3. Evaluate the inside first: $f(3) = 3^2 + 1 = 10$. Then double the output: $2 \cdot 10 = 20$. $\boxed{20}$
How it differs from $f(2 \cdot 3)$: there the doubling happens to the *input*, so the function is evaluated at 6 and gives $6^2 + 1 = 37$. Different instruction, different value. An answer of 37 is exactly this error — the factor was slid inside the parentheses, which is legal for multiplication and illegal for a function name.

4. Find and fix: Leo's $f(3) = 3(4x - 1) = 33$**, for** $f(x) = 4x - 1$**.**
Solution: Leo treated f as a quantity and $f(3)$ as a product, so he distributed a factor of 3 across the rule. Two things give it away: his answer still contains the variable x before he substitutes, and the input 3 is never placed into the rule. Correct reading — the 3 replaces x:

$$\begin{aligned} f(3) &= 4(3) - 1 \\ &= 12 - 1 = 11 \end{aligned}$$

The correct value is $\boxed{11}$
(His 33 is $3(4 \cdot 3 - 1) = 3 \cdot 11$, which is the correct output multiplied by the input — the signature of this misreading.)

5. Find and fix: Sam's $f(2 + 3) = f(2) + f(3) = 28$**, for** $f(x) = x^2 + 3x$**.**
Solution: Sam borrowed the distributive property, which applies to multiplication over addition. Function notation is not multiplication, so f does not distribute across the sum in its input slot; the parentheses tell you to add *first* and then evaluate once. Correct reading:

$$\begin{aligned} f(2 + 3) &= f(5) \\ &= 5^2 + 3(5) \\ &= 25 + 15 = 40 \end{aligned}$$

The correct value is $\boxed{40}$
His 28 is $f(2) + f(3) = 10 + 18$, and the gap of 12 is the cross-term $2 \cdot 2 \cdot 3$ that squaring the sum creates — concrete evidence that $f(a + b) \neq f(a) + f(b)$ for this rule.

6. $f(x) = 5x - 12$**; find every** x **with** $f(x) = 8$**.**
Solution: $f(x) = 8$ says "the output equals 8," so replace the output with the rule and solve the resulting equation for the input:

$$\begin{aligned} 5x - 12 &= 8 \\ 5x &= 20 \\ x &= 4 \end{aligned}$$

Check: $f(4) = 5(4) - 12 = 8$ ✓. $\boxed{x = 4}$
Why it is not a division: dividing 8 by f would require f to be a number. It is a rule, so the only way in is to substitute and solve.

7. $h(x) = x^2 - 4x$**; find every** x **with** $h(x) = 0$**.**
Solution: Substitute the rule for the output and factor:

$$x^2 - 4x = 0$$
$$x(x - 4) = 0$$

The zero-product property gives $x = 0$ or $x - 4 = 0$.

$$\boxed{x = 0 \quad \text{or} \quad x = 4}$$

What the "h times x" reading misses: reading $h(x) = 0$ as a product $h \cdot x = 0$ makes $x = 0$ look like the whole answer, because the other factor is imagined to be the letter h rather than the expression $x - 4$. Testing the missed root settles it: $h(4) = 16 - 16 = 0$, so 4 is genuinely an input with output zero.

8. Explain: telling the two readings of $f(x)$ **apart.** *(Open response — flagged for manual review; a model answer follows.)*
Model answer: The parentheses in $f(x)$ hold an *input*, not a factor. f is the name of a rule, not a quantity, so it has no numeric value to be multiplied by anything — writing $f \cdot x$ would be multiplying by a name. Substituting is the only legal move: $f(3)$ says "put 3 wherever x appears in the rule."
A concrete test a classmate can apply: evaluate the same expression both ways and compare. For $f(x) = x+4$, the function reading gives $f(3) = 3+4 = 7$, while the multiplication reading gives $3(x+4) = 21$ at $x = 3$. One number, two readings — and only the substitution reading uses the 3 as an input. A second, quicker test: if your work still contains an x after you were asked for $f(3)$, you did not substitute, so you did not evaluate the function.
What a reviewer should look for: the response must separate the input slot from a factor and must not justify itself with a case where the two readings happen to agree numerically.